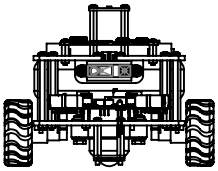
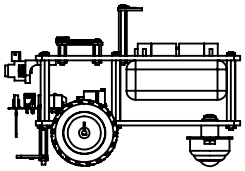
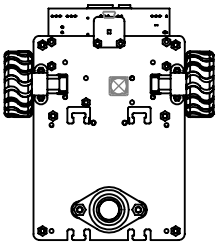
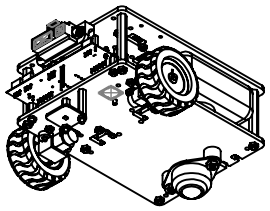
PART NUMBER	ITEM NO.	DESCRIPTION	QTY.
ASM-Rover-Layer1	1	Mounting for motors, DCDC, MotorPower Board, and DFRobotics caster	1
RoverL2-Breadboard	2	RoverL2-Breadboard	1
AliExpress 1005006038666758	3	M3 x 40 Standoff, Nylon	6
M3 x 8 Screw, Nylon	4	Pan head cross recess M3 x 8 screw, Nylon	6
M3 Nut, Nylon	5	M3 Hex thin nut, Nylon	6









UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO. ASM-Rover

A4

WEIGHT:

SCALE:1:5

SHEET 1 OF 4

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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	ASM-Rover-Layer1	Mounting for motors, DCDC, MotorPower Board, and DFRobotics caster	1
1.1	baseplate-marcel	Lasercut Acrylic 3mm	1
1.2	ASM-WheelMotor	Tire, Wheel, and Motor module	2
1.2.1	AliExpress 1005006213216702	Micro Metal Geared motor w/Encoder - 12V 300RPM 50:1	1
1.2.2	AliExpress 1005006048876448	Micro Metal Gearmotor Extended Bracket (sold as pair)	1
1.2.3	ISO - 4036 - M2 - N	M2 Hex thin nut, steel	2
1.2.4	AliExpress 1005006224428654	Wheel and Tire 42x19mm (Sold as a pair)	1
1.3	ASM-caster	Caster Assembly (DFrobotics)	1
1.3.1	DFRobots 225	19mm ball caster	1
1.3.2	spacer-caster	7.9mm 3D Print FDM (PLA)	2
1.3.3	M4 Lockwasher	Springsteel	2
1.3.4	ISO - 4036 - M4 - N	M4 Hex thin nut, steel	2
1.4	ASM-DCDC	Module to convert 7.4V to 5V	1
1.4.1	AliExpress 1005008604867476	ADIY LM2596 Step Down DC-DC Buck Converter Adjustable Module	1
1.4.2	AliExpress 1005006038666758	M2.5 x 8 Standoff, Nylon	2
1.4.3	M2.5 x 5 Screw, Nylon	Pan head cross recess M2,5 x 5 screw, Nylon	2
1.5	ISO 10669-4-N	M4 plain washer, steel	2
1.6	ASM-MotorPower	Module to distribute and control power. Motor driver JST-ZH interface to motors.	1
1.6.1	RoverMotorPower	PCB with Battery, DCDC, Motor, and IR sensor connectors	1
1.6.2	AliExpress 1005006038666758	M2.5 x 8 Standoff, Nylon	4
1.6.3	ISO 7045 - M2.5 x 5 - Z - 5N	Pan head cross recess M2,5 x 5 screw, Nylon	4
1.7	ASM-ColorSensor	Module to measure line color and offset	1
1.7.1	AliExpress 1005006745936264	RGB Color Sensor TCS34725 with IR Filter and White LED	1
1.7.2	AliExpress 1005006038666758	M2.5 x 20 Standoff, Nylon	2
1.7.3	M2.5 Nut, Nylon	M2.5 Hex thin nut, Nylon	2
1.8	ISO 7045 - M2 x 8 - Z - 8N	M2 x 8 Screw, Steel	4
1.9	M2.5 Nut, Nylon	M2.5 Hex thin nut, Nylon	6
1.10	M2.5 x 5 Screw, Nylon	Pan head cross recess M2,5 x 5 screw, Nylon	2
1.11	ISO 7045 - M4 x 10 - Z - 10N	M4 x 10 Screw, Steel	2
2	RoverL2-Breadboard	RoverL2-Breadboard	1
2.1	baseplate-marcel	Lasercut Acrylic 3mm	1
2.2	C1020	10000mAh Power Bank with USB-C and 7.4V DC jack	1
2.3	ASM-IMU	Module to measure position, heading, and temperature	1
2.3.1	AliExpress 1005005499263696	ICM-20948 Sensor Module 9 Axis MEMS Motion Tracking Device Sensor	1
2.3.2	AliExpress 1005006038666758	M2.5 x 30 Standoff, Nylon	2
2.3.3	M2.5 x 5 Screw, Nylon	Pan head cross recess M2,5 x 5 screw, Nylon	2
2.4	M2.5 Nut, Nylon	M2.5 Hex thin nut, Nylon	8
2.5	ASM-MCU	MicroControl Unit Module	1
2.5.1	PiHut 43855634497731	Raspberry Pi Zero 2 W or 1 W	1
2.5.2	AliExpress 1005006038666758	M2.5 x 8 Standoff, Nylon	4
2.5.3	M2.5 x 5 Screw, Nylon	Pan head cross recess M2,5 x 5 screw, Nylon	4
2.6	ASM-DistanceSensor	Measure distance to obstacle including other Rovers	1
2.6.1	bracket-sharp	90-degree mounting bracket with M2.5 holes	1
2.6.2	AliExpress 1005006745936264	Sharp GP2Y0A21YK0F 10-80cm infrared distance sensor	1
2.6.3	M2.5 Nut, Nylon	M2.5 Hex thin nut, Nylon	2
2.6.4	M2.5 x 5 Screw, Nylon	Pan head cross recess M2,5 x 5 screw, Nylon	2
2.7	M2.5 x 5 Screw, Nylon	Pan head cross recess M2,5 x 5 screw, Nylon	2
2.8	AliExpress 1005005995135021	30-row 2x5 column half-size breadboard	1
3	AliExpress 1005006038666758	M3 x 40 Standoff, Nylon	6
4	M3 x 8 Screw, Nylon	Pan head cross recess M3 x 8 screw, Nylon	6
5	M3 Nut, Nylon	M3 Hex thin nut, Nylon	6

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
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DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

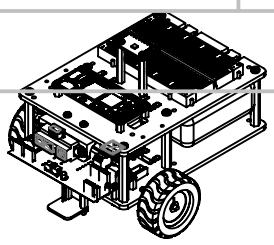
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DWG NO. ASM-Rover

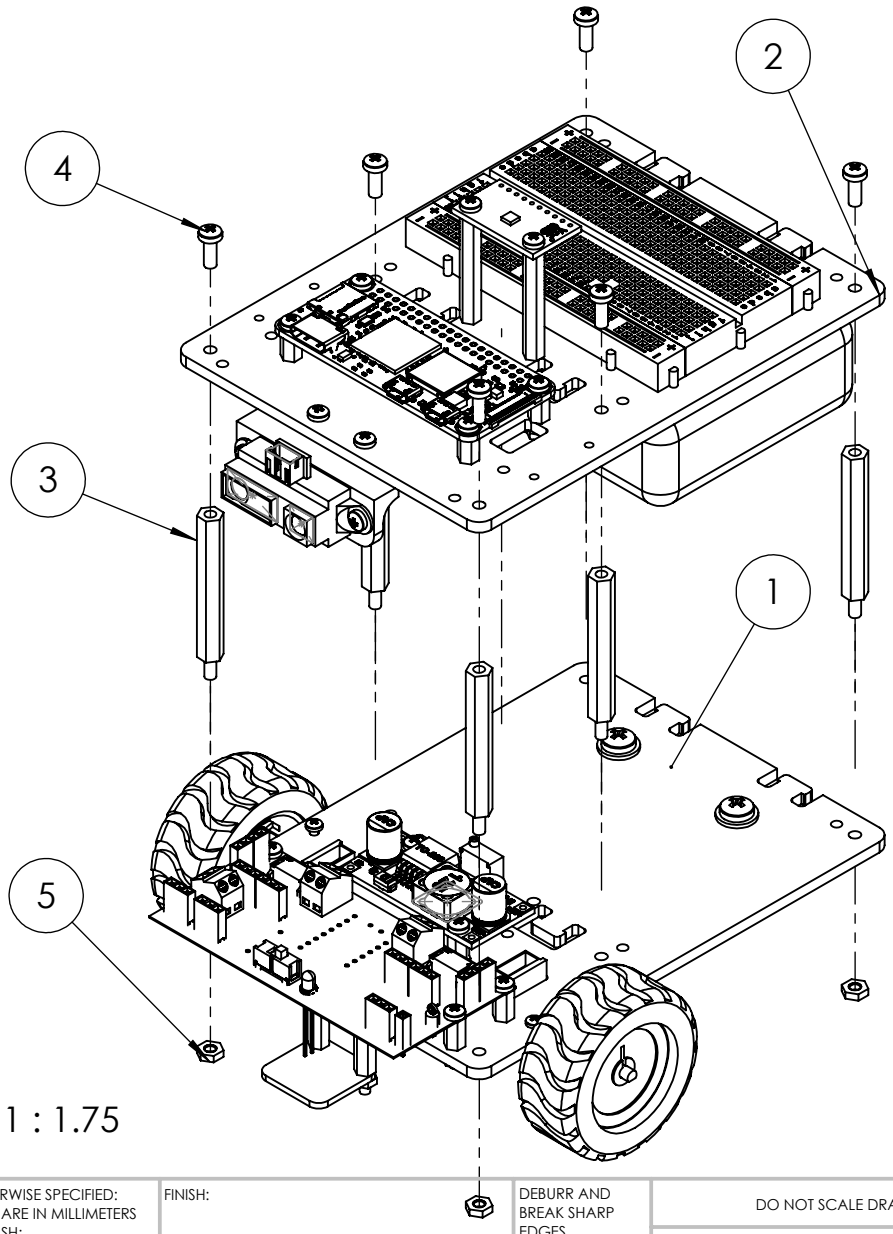
A4

SCALE:1:5

SHEET 2 OF 4

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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	ASM-Rover-Layer1	Mounting for motors, DCDC, MotorPower Board, and DFRobotics caster	1
2	RoverL2-Breadboard	RoverL2-Breadboard	1
3	AliExpress 1005006038666758	M3 x 40 Standoff, Nylon	6
4	M3 x 8 Screw, Nylon	Pan head cross recess M3 x 8 screw, Nylon	6
5	M3 Nut, Nylon	M3 Hex thin nut, Nylon	6

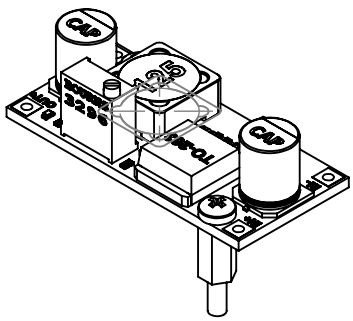


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UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
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APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. ASM-Rover	A4
				WEIGHT:		SCALE:1:5	SHEET 3 OF 4

SOLIDWORKS Educational Product. For Instructional Use Only.

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	AliExpress 1005008604867476	ADIY LM2596 Step Down DC-DC Buck Converter Adjustable Module	1
2	AliExpress 1005006038666758	M2.5 x 8 Standoff, Nylon	2
3	M2.5 x 5 Screw, Nylon	Pan head cross recess M2,5 x 5 screw, Nylon	2



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
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CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. ASM-Rover	A4
				WEIGHT:		SCALE:1:1	SHEET 4 OF 4

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